

# AgriConnect — Software Requirements Specification

## Module: Farmer Role

**Document version:** 1.0 **Prepared for:** Development Team **Status:** Draft for review

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### 1. Purpose

This document specifies the functional and non-functional requirements for the **Farmer role** of the AgriConnect platform. It is intended to give the development team enough detail to design the database schema, build the API, and implement the UI screens without needing further clarification on core behavior.

AgriConnect is a **0% mediator marketplace**: it only connects farmers and buyers. It does not handle negotiation, logistics, payment for goods, or quality assurance. This principle drives several requirements below (e.g., no accept/reject step on bookings, no dispute resolution for the trade itself).

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### 2. Scope

This document covers only the **Farmer role**: registration, profile, product management, booking visibility, communication unlock, and safety reporting. The Buyer role, Admin role, and payment gateway integration are separate documents and are referenced here only where they intersect with Farmer workflows.

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### 3. Definitions and Glossary

| Term    | Meaning                                                                                 |
|---------|-----------------------------------------------------------------------------------------|
| Farmer  | A registered user who lists agricultural products for sale                              |
| Buyer   | A registered user who browses and books products                                        |
| Booking | A record created when a buyer pays the booking fee to unlock a farmer's contact details |

|                |                                                                                                                     |
|----------------|---------------------------------------------------------------------------------------------------------------------|
| Booking Fee    | A small, non-refundable-by-default platform fee paid by the buyer to unlock contact details (not a product deposit) |
| Listing        | A product entry created by a farmer                                                                                 |
| Contact Unlock | The event where farmer/buyer contact details become mutually visible after a booking                                |
| OTP            | One-Time Password used for mobile verification                                                                      |

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## 4. Actors

| Actor  | Description                                                  |
|--------|--------------------------------------------------------------|
| Farmer | Primary actor for this document                              |
| Buyer  | Secondary actor — triggers booking notifications and unlocks |
| Admin  | Secondary actor — reviews reports, can flag/remove listings  |
| System | Background actor — handles OTP, expiry jobs, notifications   |

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## 5. Farmer Role Overview

The Farmer creates an account, completes a profile, lists products, and is notified when a buyer books a product. The farmer does not transact through the platform — AgriConnect's involvement ends once contact details are exchanged.

### High-level flow:

Register → Verify OTP → Complete Profile → Dashboard → Add Product → Publish → Receive Booking → View Buyer Contact → Communicate Directly → Update Product Status

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## 6. Data Model (Entities)

Providing this up front saves the developer from reverse-engineering the schema out of the workflow text.

### 6.1 Farmer

| Field | Type | Required | Notes |
|-------|------|----------|-------|
|-------|------|----------|-------|

|                    |                |     |                                             |
|--------------------|----------------|-----|---------------------------------------------|
| farmer_id          | UUID/PK        | Yes | System-generated                            |
| full_name          | string         | Yes | Max 100 chars                               |
| mobile_number      | string(10-15 ) | Yes | Unique per role; E.164 or India-only format |
| mobile_verified    | boolean        | Yes | Default false until OTP success             |
| email              | string         | No  | Validated format if provided                |
| password_hash      | string         | Yes | Never store plaintext                       |
| profile_photo_url  | string         | No  |                                             |
| alternate_mobile   | string         | No  |                                             |
| state              | string         | Yes |                                             |
| district           | string         | Yes |                                             |
| mandal_taluk       | string         | No  |                                             |
| village            | string         | Yes | Hidden from buyers pre-booking              |
| preferred_language | enum           | Yes | e.g., en, hi, te, ta...                     |
| account_status     | enum           | Yes | ACTIVE, SUSPENDED, DELETED                  |
| created_at         | datetime       | Yes |                                             |
| updated_at         | datetime       | Yes |                                             |

**Business rule:** `mobile_number` must be unique **within the FARMER role**. Whether the same number may also register as a Buyer is a product decision the team should confirm — see Open Question 9.1.

## 6.2 Product (Listing)

| Field        | Type       | Required | Notes                                         |
|--------------|------------|----------|-----------------------------------------------|
| product_id   | UUID/PK    | Yes      |                                               |
| farmer_id    | FK         | Yes      |                                               |
| product_name | string     | Yes      | Max 100 chars                                 |
| category     | enum       | Yes      | Vegetable, Fruit, Grain, Pulses, Spice, Other |
| images       | array<url> | Yes      | Min 1, max 8 recommended                      |

|                          |              |             |                                                       |
|--------------------------|--------------|-------------|-------------------------------------------------------|
| quantity_available       | decimal      | Yes         | Must be > 0                                           |
| unit                     | enum         | Yes         | kg, quintal, ton, bag, dozen                          |
| expected_price           | decimal      | No          |                                                       |
| price_unit               | enum         | Conditional | Required if expected_price is set                     |
| harvest_date             | date         | Yes         | Cannot be more than X days in the past (configurable) |
| description              | text         | No          | Max ~1000 chars                                       |
| state / district         | string       | Yes         | Shown publicly                                        |
| village / exact_location | string / geo | Yes         | <b>Hidden until booking</b>                           |
| status                   | enum         | Yes         | DRAFT, ACTIVE, RESERVED, SOLD, EXPIRED, REMOVED       |
| listing_expiry_date      | date         | Yes         | Auto-set (e.g., created_at + 30 days), configurable   |
| created_at / updated_at  | datetime     | Yes         |                                                       |

### 6.3 Booking

| Field                | Type     | Required | Notes                                |
|----------------------|----------|----------|--------------------------------------|
| booking_id           | UUID/PK  | Yes      |                                      |
| product_id           | FK       | Yes      |                                      |
| buyer_id             | FK       | Yes      |                                      |
| farmer_id            | FK       | Yes      | Denormalized for query speed         |
| quantity_of_interest | decimal  | No       | Optional, informational only         |
| booking_fee_amount   | decimal  | Yes      |                                      |
| payment_status       | enum     | Yes      | PENDING, PAID, FAILED, REFUNDED      |
| booking_status       | enum     | Yes      | CONFIRMED, COMPLETED, CANCELLED      |
| contact_unlocked_at  | datetime | Yes      | Set the moment payment_status = PAID |

|            |          |     |
|------------|----------|-----|
| created_at | datetime | Yes |
|------------|----------|-----|

## 6.4 Report

| Field       | Type           | Required | Notes                                            |
|-------------|----------------|----------|--------------------------------------------------|
| report_id   | UUID/PK        | Yes      |                                                  |
| booking_id  | FK             | Yes      | Reports are tied to a booking, not freeform      |
| reported_by | FK (farmer_id) | Yes      |                                                  |
| reason      | enum           | Yes      | FAKE_BUYER, SPAM, ABUSIVE, FRAUD, NO_SHOW, OTHER |
| description | text           | No       | Max 500 chars                                    |
| status      | enum           | Yes      | OPEN, UNDER_REVIEW, RESOLVED, DISMISSED          |
| created_at  | datetime       | Yes      |                                                  |

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## 7. Farmer Registration Workflow

### 7.1 Required Fields

| Field                      | Required | Validation                                           |
|----------------------------|----------|------------------------------------------------------|
| Full Name                  | Yes      | 2–100 chars, letters/spaces only                     |
| Mobile Number              | Yes      | 10-digit Indian mobile format, unique                |
| OTP Verification           | Yes      | 6-digit, expires in 5 minutes, max 3 resend attempts |
| Email                      | No       | Standard email regex if provided                     |
| Password                   | Yes      | Min 8 chars, at least 1 letter + 1 number            |
| Confirm Password           | Yes      | Must match Password                                  |
| State / District / Village | Yes      | From a predefined location dataset (not free text)   |
| Preferred Language         | Yes      | From supported language list                         |

### 7.2 Functional Requirements

- The system shall prevent duplicate Farmer accounts on the same verified mobile number.
- The system shall send an OTP via SMS and expire it after 5 minutes.
- The system shall lock OTP requests after 5 failed attempts for 15 minutes (basic brute-force protection).
- The system shall hash passwords (bcrypt/argon2) — never store or log plaintext.
- The system shall reject registration if location fields don't match the supported state/district dataset.
- The system shall assign role **FARMER** only after successful OTP verification.

### 7.3 Flow

Open App → Select "Register as Farmer" → Enter Details → Receive & Enter OTP → Account Created → Redirect to Complete Profile

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## 8. Farmer Login Workflow

- Login via **mobile + password** or **mobile + OTP**.
  - The system shall lock the account for 15 minutes after 5 consecutive failed password attempts.
  - The system shall reject login for **SUSPENDED** or **DELETED** accounts with a clear message.
  - The system shall provide OTP-based password reset (no security questions).
  - Session shall use a signed token (JWT or equivalent) with a reasonable expiry (e.g., 7 days) and refresh mechanism.
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## 9. Farmer Profile Workflow

### 9.1 Privacy Rule (Critical)

Before a booking is confirmed for a given buyer, that buyer must **never** receive, in any API response, the following fields: **full\_name**, **mobile\_number**, **alternate\_mobile**, **village**, **exact\_location**. This must be enforced at the **API/serialization layer**, not just hidden in the UI — a buyer inspecting network traffic must not be able to see it either.

After **booking\_status** = **CONFIRMED** for a specific (**product\_id**, **buyer\_id**) pair, that specific buyer's session may retrieve the farmer's permitted fields. Other buyers who haven't booked still cannot.

### 9.2 Editable Fields

Name, profile photo, alternate number, location, preferred language, password. Changing the **primary verified mobile number** requires a fresh OTP verification and should not simply overwrite the old number until the new one is confirmed.

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## 10. Farmer Dashboard Workflow

Summary counters: Total Products, Active, Reserved, Sold, Total Booking Requests, Recent Activity feed.

Navigation: Home, Add Product, My Products, Bookings, Notifications, Profile.

**Non-functional note:** dashboard counts should be served from a cached/aggregated source if the farmer has a large number of listings, rather than computed live on every page load.

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## 11. Add Product Workflow

### 11.1 Fields — see §6.2 for full schema. Additional validation rules:

- `quantity_available` must be a positive number, max 2 decimal places.
- `expected_price`, if provided, must be positive; `price_unit` becomes required the moment `expected_price` is set.
- `harvest_date` must not be more than 7 days in the past (configurable) and not more than 180 days in the future.
- Minimum 1 image, maximum 8. Accepted formats: JPG, PNG, WEBP. Max file size 5MB each (compress server-side to a max dimension, e.g., 1600px).
- `village` / exact location is stored but must be excluded from any public/buyer-facing GET response pre-booking (see §9.1).

### 11.2 Flow

Dashboard → Add Product → Enter Info → Upload Images → Preview → Submit → Server Validation → Status = ACTIVE

### 11.3 Functional Requirements

- The system shall assign a unique `product_id` and timestamp on creation.
- The system shall reject submission if any mandatory field is missing, returning field-level errors (not a generic failure).
- The system shall auto-set `status = DRAFT` if the farmer exits before submitting, and `ACTIVE` upon successful publish.

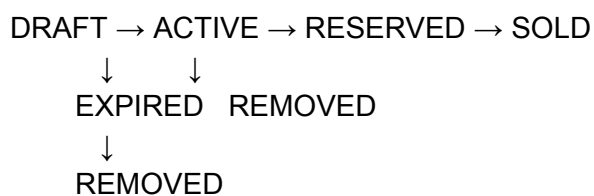
- The system shall set a `listing_expiry_date` automatically and notify the farmer 3 days before expiry.

## 12. Product Status Workflow

| Status   | Meaning                           | Can transition to                                      |
|----------|-----------------------------------|--------------------------------------------------------|
| DRAFT    | Saved, not public                 | ACTIVE, REMOVED                                        |
| ACTIVE   | Publicly visible, bookable        | RESERVED, SOLD, EXPIRED, REMOVED                       |
| RESERVED | Temporarily held by farmer        | ACTIVE, SOLD, REMOVED                                  |
| SOLD     | No longer available               | (terminal, or ACTIVE if farmer reopens with new stock) |
| EXPIRED  | Past validity, auto-set by system | ACTIVE (if farmer renews), REMOVED                     |
| REMOVED  | Deleted by farmer or admin        | (terminal)                                             |

**Functional requirement:** the system shall reject new bookings for any product not in `ACTIVE` status, with a clear "no longer available" message rather than a silent failure.

**State diagram (text form):**



## 13. My Products Workflow

Each product card: image, name, quantity, location, upload date, status, booking count.

Farmer actions: View, Edit, Update Quantity/Price, Manage Images, Change Status, Delete.

**Business rule:** if a product has an active (unconfirmed-as-complete) booking, the system shall show a confirmation warning before allowing deletion or a status change to REMOVED/SOLD, e.g.:

"This product has an active booking. Changing its status may affect an interested buyer. Continue?"

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## 14. Booking Notification Workflow

On successful buyer payment (`payment_status = PAID`):

1. Booking record created with `booking_status = CONFIRMED`.
2. `contact_unlocked_at` timestamp set immediately.
3. Push/SMS notification sent to farmer.
4. Farmer's permitted contact fields become visible to that buyer only.
5. Buyer's permitted contact fields become visible to the farmer.

**Design decision (carried over intentionally):** there is **no** Accept/Reject step. Since the buyer has already paid, contact unlock is automatic. This keeps the 0% mediator promise consistent — see §17 open question on refund policy, which is the one place this needs a documented rule.

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## 15. Farmer Booking Management Workflow

Tabs: New, Active, Completed, Cancelled (if supported).

Per booking, farmer sees: Booking ID, Product, Product Image, Buyer Name & Phone, Booking Date, Quantity of Interest, Status.

Actions: Call Buyer, Message Buyer, View Details, Mark Deal Completed, Mark Product Sold, Report Buyer.

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## 16. Direct Communication Workflow

Actions: Call, WhatsApp deep-link (`https://wa.me/<number>`), Copy Number.

Platform explicitly does **not**: negotiate price, guarantee the transaction, arrange transport, collect payment for goods, judge quality, or act as an agent.

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## 17. Product Completion Workflow

Farmer updates listing post-transaction: Still Available, Partially Available (enter remaining quantity), Reserved, Sold Out.

Example: 1,000 kg listed → 600 kg sold → farmer enters 400 kg remaining → listing stays live at reduced quantity.

**Sold Out** → system shall block further bookings on that product immediately.

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## 18. Notifications Workflow

Trigger events: new booking, listing expiring in 3 days, listing expired, admin flagged/removed a listing, security events (password change, new device login).

Delivery channels: in-app + SMS/push for time-sensitive events (new booking, security). In-app only is acceptable for informational events (listing expiring).

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## 19. Report and Safety Workflow

Reasons: Fake Buyer, Spam, Abusive Behavior, Fraud Attempt, No-show, Other.

**Booking Details** → **Report Buyer** → **Select Reason** → **Optional Description** → **Submit** → **Admin Queue**

**Functional requirement:** the system shall rate-limit reports (e.g., max 10 per farmer per day) to prevent abuse of the reporting system itself, and shall notify the farmer once a report status changes.

This is a platform safety feature only — it does not make AgriConnect a party to the underlying trade dispute.

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## 20. Complete Farmer Journey (End-to-End)

**Register** → **Verify OTP** → **Complete Profile** → **Dashboard** → **Add Product** → **Upload Images/Details** → **Preview** → **Publish** → **Product Visible to Buyers** → **Buyer Pays Booking Fee** → **Booking Confirmed** → **Farmer Notified** → **Contact Details Unlocked** → **Direct Communication** → **Farm Visit & Negotiation (off-platform)** → **Farmer Updates Quantity/Status**

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## 21. Functional Requirements Summary

| ID       | Requirement                                                                                                                                                                        |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-F-001 | The system shall allow a user to register as a Farmer.                                                                                                                             |
| FR-F-002 | The system shall verify the farmer's mobile number via OTP before activating the account.                                                                                          |
| FR-F-003 | The system shall allow the farmer to create and update a profile.                                                                                                                  |
| FR-F-004 | The system shall allow the farmer to add agricultural products with the fields defined in §6.2.                                                                                    |
| FR-F-005 | The system shall support 1–8 images per product with server-side compression.                                                                                                      |
| FR-F-006 | The system shall allow the farmer to edit, update, and remove listings.                                                                                                            |
| FR-F-007 | The system shall enforce, at the API layer, that farmer contact and exact-location fields are excluded from responses to buyers who have not completed a booking for that product. |
| FR-F-008 | The system shall notify the farmer immediately upon a successful buyer booking.                                                                                                    |
| FR-F-009 | The system shall reveal buyer contact information to the farmer only after <code>payment_status = PAID</code> .                                                                    |
| FR-F-010 | The system shall allow the farmer to update remaining quantity and availability status.                                                                                            |
| FR-F-011 | The system shall allow the farmer to mark a product as sold, blocking further bookings.                                                                                            |
| FR-F-012 | The system shall maintain a full, timestamped booking history per farmer.                                                                                                          |
| FR-F-013 | The system shall allow farmers to report a buyer tied to a specific booking, with rate limiting.                                                                                   |
| FR-F-014 | The system shall not participate in negotiation, logistics, transportation, or final payment for goods.                                                                            |
| FR-F-015 | The system shall reject new bookings on products with status other than ACTIVE.                                                                                                    |

- FR-F-016 The system shall auto-expire listings after a configurable period and notify the farmer in advance.
- FR-F-017 The system shall warn the farmer before deleting or changing status on a product with an active booking.
- FR-F-018 The system shall require re-verification via OTP before changing the primary mobile number.
- FR-F-019 The system shall log all status transitions on products and bookings for audit purposes.
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## 22. Non-Functional Requirements

| Category      | Requirement                                                                                                                                                                                                                      |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Security      | Passwords hashed (bcrypt/argon2); all traffic over HTTPS; OTP rate-limited; JWT/session tokens signed and expiring                                                                                                               |
| Privacy       | Contact/exact-location fields excluded server-side pre-booking, not just hidden in UI (see FR-F-007)                                                                                                                             |
| Performance   | Dashboard and product listing queries should respond within 1–2 seconds under normal load; image uploads compressed asynchronously where possible                                                                                |
| Scalability   | Product and booking tables should be indexed on <code>farmer_id</code> , <code>status</code> , and <code>district</code> for filtered browsing at scale                                                                          |
| Availability  | Target 99.5%+ uptime; OTP and notification delivery should degrade gracefully (retry queue) if SMS gateway is down                                                                                                               |
| Localization  | UI and notification templates must support the configured language list (§7.1); location data should not be free text to keep filtering reliable                                                                                 |
| Auditability  | All product status changes and booking state changes shall be logged with timestamp and actor                                                                                                                                    |
| Accessibility | Forms should support screen readers and reasonable contrast ratios, given that many farmer users may have low digital literacy — favor large touch targets and minimal required typing (dropdowns over free text where possible) |

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## 23. Error Handling & Edge Cases

Developers should explicitly handle these — they're easy to miss in a happy-path build:

1. **Simultaneous bookings on limited stock:** two buyers booking the last unit at once. Decide: is booking fee tied to a reserved quantity, or is it purely "unlock contact," with stock conflicts resolved off-platform between farmer and buyer? (Recommendation: keep it as contact-unlock only, since AgriConnect doesn't manage stock reservation — but this should be an explicit, documented decision, not an accident.)
  2. **Buyer pays but farmer never responds:** define an SLA-free but visible expectation, e.g., booking shows "awaiting farmer response" after 48 hours, without triggering a refund obligation (see open question 23.1 below on refund policy).
  3. **Farmer deletes a product with a pending booking:** block or warn per §13 — don't allow silent deletion.
  4. **OTP delivery failure:** provide a resend option and a fallback support contact path.
  5. **Duplicate product images across listings:** not a blocker, but consider basic duplicate-image detection to reduce spam/fake listings.
  6. **Listing expiry while a booking is active:** an active booking should not auto-expire the listing out from under an in-progress deal; expiry logic should check for open bookings first.
  7. **Farmer account suspended mid-booking:** buyers with already-unlocked contact info should retain that info; suspension should block *new* activity, not retroactively hide completed unlocks.
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## 24. Open Questions for Product Owner (Resolve Before Development)

| #    | Question                                                                                                                                                                                            |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 24.1 | Is the booking fee refundable under any circumstance (e.g., farmer never responds, listing was fraudulent)? This needs an explicit policy even though AgriConnect doesn't mediate the trade itself. |
| 24.2 | Can the same mobile number be registered as both Farmer and Buyer, or are the roles mutually exclusive per number?                                                                                  |
| 24.3 | What is the default listing expiry period (e.g., 30 days), and is it configurable per category (e.g., perishables vs. grain)?                                                                       |
| 24.4 | Does a booking reserve a specific quantity, or is it purely a contact-unlock event unconnected to stock? This affects the data model and the "simultaneous booking" edge case in §23.               |
| 24.5 | Is there a cap on how many active bookings a single buyer can hold at once, to prevent scraping farmer contact info at scale?                                                                       |

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## 25. Traceability Note

Each FR-F-#### ID in §21 should be referenced directly in test cases and, if used, in the project's issue tracker, so QA can trace coverage back to this document.